

Honors Class04 Activity: Working with Data Tables

Team Members: _____

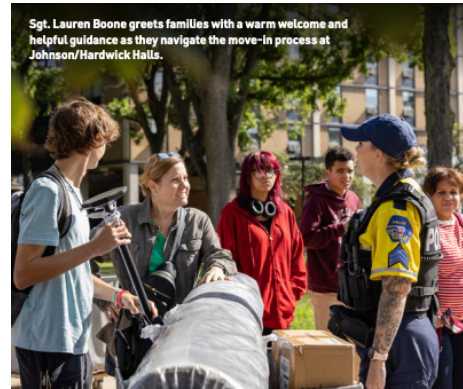
Learning Objectives

- Introduction to the `datascience` module
- Learn how to create and manipulate data tables

Data Tables

Tabular data is common in all fields, including science, so we need ways to work with data tables in Python. We will be using the *datascience* module in this course.

Our first table we will build from scratch. In Lab03 you will save typing by reading data from a file into a table. The data comes from Temple's 2025 Annual Security and Fire Safety Report. To keep the table small, we will focus on just a small subset of the data – crime in the residence halls and crime categories with non-zero values. For example, there was no embezzlement, vagrancy, or driving under the influence (surprise!) in the dorms, so those rows have been omitted.



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Open a fresh Jupyter notebook and enter the following in a code cell.

```
# Load all of the functions in the datascience module
from datascience import *
```

```
# Data for each column goes in a list or array.
```

```
categories = [
    "Simple Assault",
    "Harassment",
    "Fraud",
    "Theft",
    "Vandalism",
    "Disorderly Conduct",
    "Sex Assault (Other)",
    "Drunkenness",
    "All Other Offenses",
]
```

```
counts_2024 = [13, 44, 6, 30, 13, 1, 1, 8, 0]
counts_2023 = [0, 61, 9, 24, 17, 6, 0, 7, 1]
counts_2022 = [3, 32, 9, 23, 9, 0, 1, 5, 6]
```

```
# Create the table
```

```
res_hall = Table().with_columns(
    "Category", categories,
```

```
"2024", counts_2024,  
"2023", counts_2023,  
"2022", counts_2022,  
)  
  
res_hall
```

It is important to understand what just happened. A data table is comprised of columns. The columns are arrays of data. Different columns can hold different types of data, but all of the rows in a given column must be the same data type. Here we have one column of strings (the type of crime) and three numeric columns for the three years of data. Each column in the table has a label — the column header.

So to build this table, we first created a list for each column. Then we called `Table().with_columns()` to build the table. The arguments to this function were label, array, label, array, ... for as many columns as needed. Note: all of the columns must have the same number of rows.

Working with Data Tables

Once the data is in a `datascience` table, you have many properties and methods available to operate on the data.

Table Properties

Table properties are just that — properties of the table. They do not involve manipulating the data. Here are a couple of examples for you to try. **Type them into your notebook.**

```
# The number of rows is a table property  
res_hall.num_rows
```

```
# The number of columns is a table property  
res_hall.num_columns
```

```
# The collection of column headers is a table property  
res_hall.labels
```

Table Methods

Unlike properties, table methods are Python functions built into data tables that operate on the table data. **Type the following examples into your notebook and see if you can figure out what they do.**

```
res_hall.select('Category', '2024')
```

The `select()` method does what?

```
res_hall.column('2024')
```

The `column()` method does what?

```
res_hall.take(3)
```

The `take()` method does what?

```
res_hall.sort('2024')
```

The `sort()` method does what?

Filtering Tables

In Lab03 you will get practice with the `where()` method, which returns the rows of a table that meet some condition specified in the `where predicate`. You pass the `where()` function the label for the column you want to filter and the condition. Here we return a new table with just the rows where the `2024` column has values greater than 25.

```
res_hall.where('2024', are.above(25))
```

```
res_hall.where("Category", are.equal_to("Fraud"))
```

(One wonders what sort of fraud occurs in residence halls.)

```
res_hall.where("2022", are.between(5, 10))
```

Suppose we wanted to know the total number of incidents in the table for 2024. One way to do this is to extract the data from that column and sum it.

```
incidents_2024 = res_hall.column("2024")  
sum(incidents_2024)
```

How many total incidents were there in 2022?

For a small table such as this one, it might be easier to use a calculator, but imagine a table with millions of rows!

Discussion Questions

The activity emphasizes that columns are arrays.

Why do you think the designers of the `datascience` module emphasize column-wise thinking instead of row-wise thinking? How does this choice influence the kinds of questions that are easy vs. hard to ask?

The `where()` method filters rows based on conditions.

Is filtering a neutral operation? How could filtering choices unintentionally bias an analysis?

Suppose `sum(res_hall.column("2024"))` returned a value that surprised you.

What would be your first step in diagnosing the issue? Which intermediate checks would you perform before assuming the data were wrong?

The activity computes total incidents by summing a column.

What assumptions are required for this total to be meaningful? Are all incidents equally “countable” in the same way?